

B1GMB42 Motherboard Slot & Port Inventory

Machine: Dell Precision Tower 5810

Serial: B1GMB42 (same chassis; DMI serial blank until SMBIOS repopulated)

Hostname: Tower5810

Motherboard: Dell 0K240Y (BIOS **A34**)

Source: live scan 2026-07-03 (fresh reinstall, **no encryption**)

CPU: Intel Xeon E5-1660 v3 @ 3.00 GHz (1 socket, 16 threads)

Regulatory: D01T / D01T006 (Dell Precision Tower 5810)

Dell sources used for this section: Owner's Manual (precision-t5810-om-pub-en-us.pdf), Quick Setup Guide, Serial PCIe Setup Guide, Dual-Host Card WP. Diagrams extracted to docs/images/.

Reference diagrams (Dell Owner's Manual)

Labels and quick-reference tables are **left**; Dell figures are **right** (PDF layout only via LaTeX). Source: precision-t5810-om-pub-en-us.pdf.

Figure 1 — Front & Rear I/O

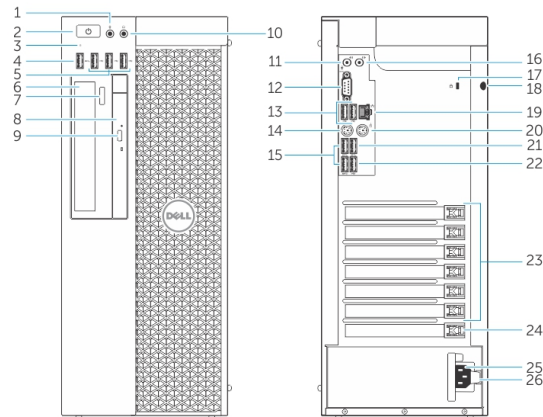
Every external user-facing port.

Front (1-11)

- 1 Mic-in
- 4 USB 3.0
- 5-6 USB 2.0 × 2
- 6-9 Slim ODD bays (opt.)
- 10 Headphone
- 11 Line-in / mic

Rear (12-26)

- 12 Serial DB-9
- 13 USB 2.0 × 2
- 14-20 PS/2, USB3, audio, RJ-45
- 23 **GPU + PERC brackets**
- 25 AC inlet (IEC C14)



B1GMB42: 12 mini-DP on 3 FirePro; Intel I217-LM RJ-45.

Figure 2 — Inside (top view)

Drive cages, cooling, first GPU.

- 1 CPU heatsink + fan → CPU_FAN
- 2 Memory shrouds (8 DIMM)
- 4 5.25" + slim ODD bays
- 5 **HDD cages** HDD0-3
- 6 PSU cable shroud
- 7 Graphics cards Slot1/2/4
- 8 Intrusion switch

B1GMB42: Hitachi rpool; TEAM special; Seagate Windows.

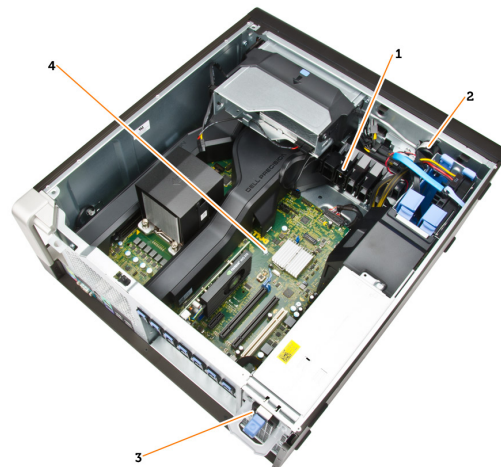


Figure 3 — Inside (PSU side)

Power module, retention cage, speaker.

- 1 PCIe retention latch
- 2 Speaker → SPK header
- 3 PSU 425/685 W
- 4 Motherboard OK240Y

Cables via shroud:

24-pin ATX (29), 10-pin CPU (30),
SATA power to each drive.

PERC H710 in Slot5.

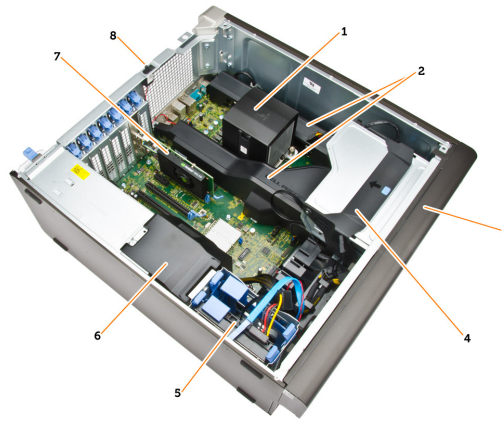
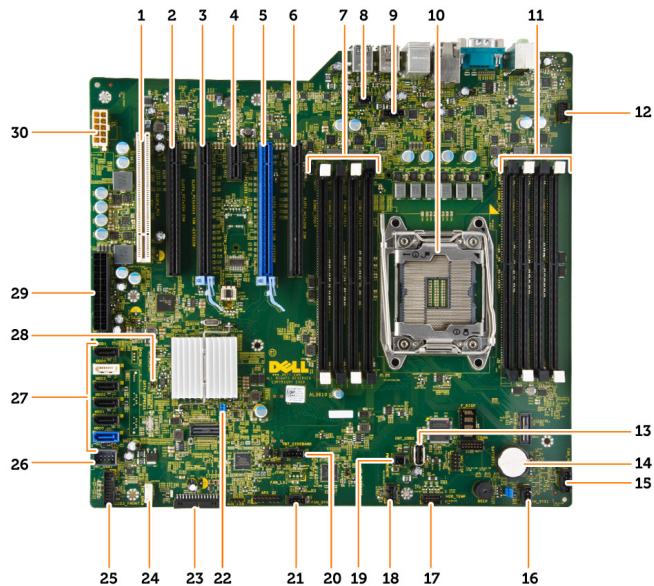


Figure 4 — System board labels (1-30)

- 1 PCI slot (Slot6)
- 2 PCIe x16 / x4 wired (Slot5)
- 3 PCIe 3.0 x16 (Slot4)
- 4 PCIe 2.0 x1 (Slot3)
- 5 PCIe 3.0 x16 (Slot2)
- 6 PCIe 3.0 x16 / x8 (Slot1)
- 7 DIMM slots (channel A)
- 8 Intrusion-switch connector
- 9 CPU heatsink fan (5-pin)
- 10 Processor socket LGA2011-3
- 11 DIMM slots (channel B)
- 12 Front-panel audio (HD)
- 13 Internal USB 2.0 header
- 14 Coin-cell battery CR2032
- 15 Optional HDD fan (5-pin)
- 16 System fan (4-pin)
- 17 HDD temperature sensor
- 18 System fan (4-pin)
- 19 PWR_REMOTE (host card)
- 20 Thunderbolt sideband
- 21 System fan (4-pin)
- 22 Password jumper
- 23 Front panel + USB 2.0
- 24 Internal speaker
- 25 USB 3.0 front-panel
- 26 Internal USB flexbay
- 27 SATA HDD0-3 + ODD0-1
- 28 RTC_RST jumper
- 29 24-pin system power
- 30 CPU power (10-pin EPS)



B1GMB42 summary (Fig 4)

Slots 1 W5000, 2 W5100, 4 W5000, 5 PERC H710, 3 empty
SATA 27 Hitachi rpool, Seagate Windows, TEAM special
Power 29 ATX + 30 CPU EPS from PSU shroud
Front 12 audio, 23 panel/LED/USB2, 25 USB3
Fans 9 CPU + 16/18/21 chassis
PCIe bus 01:00 / 02:00 / 03:00 GPUs; 07:00 PERC

Quick slot map (B1GMB42 as-built)

Chassis front

```

      |
      v
+---+-----+
|CPU| DIMM1 DIMM2 DIMM3 (empty x5)
|fan| [==== memory shrouds ====]
+---+-----+
| ODD / 5.25" bay (top) |
| HDD cage 0 | HDD cage 1 |   <- Hitachi, Seagate, TEAM
+---+-----+
| Slot1 W5000 | Slot2 W5100 |   PCIe retention cage
| Slot4 W5000 | Slot5 PERC H710 |
+---+-----+
  
```

| PSU (bottom rear) |
+-----+
Rear I/O + GPU brackets + AC inlet

Figure file	Path
Fig 1	docs/images/figure-01-front-back.jpg
Fig 2	docs/images/figure-02-inside-top.jpg
Fig 3	docs/images/figure-03-inside-psu.jpg
Fig 4	docs/images/figure-04-system-board-detail.jpg

Reinstall notes (post-TPM/disk incident)

Previous install used **ZFS native encryption + encrypted swap (TPM)**. Disk changes with TPM locked the system. This install is **identical hardware** with:

Item	Old install	This install
ZFS rpool encryption	on (TPM)	off
Swap	dm-crypt on sdb3	plain sdb3
Hostname	PT5810	Tower5810
ZFS root dataset	ubuntu_mefuj7	ubuntu_cortt9
TEAM SSD	special vdev on rpool	attached — special_small_blocks=32K
USB 2×2TB WD enclosure	present (flaky)	removed — Ventoy stick only
Apport	enabled	purged

As-Built PCIe Slots (Dell silkscreen)

Dell slot	Bus address	Electrical	Status	Attached
Slot1	01:00.0	PCIe 3 x16	In use	AMD FirePro W5000 + HDMI audio (amdgpu)
Slot2	02:00.0	PCIe 3 x16	In use	AMD FirePro W5100 + HDMI audio (amdgpu)
Slot3	—	PCIe 2	Empty	—
Slot4	03:00.0	PCIe 3 x16	In use	AMD FirePro W5000 + HDMI audio (amdgpu)
Slot5	07:00.0	PCIe 2 x4	In use	Dell PERC H710 — FW FAULT 0x40000
Slot6	—	Legacy PCI	Riser	Open

GPU device mapping

PCI	Card	DRI device	Render node
01:00.0	FirePro W5000	/dev/dri/card1	renderD128
02:00.0	FirePro W5100	/dev/dri/card2	renderD129

PCI	Card	DRI device	Render node
03:00.0	FirePro W5000	/dev/dri/card3	renderD130

All three load **amdgpu**. W5100 logs Cannot find any crtc or sizes when headless (no monitor cable).

Dell silkscreen PCIe map (factory spec)

Dell slot	Electrical (manual)	Max power (manual)	Typical use
Slot1	PCIe 3.0 x16 (wired x8)	75 W slot	Primary GPU / boot display
Slot2	PCIe 3.0 x16	75 W slot	Second GPU or accelerator
Slot3	PCIe 2.0 x1	25 W	NIC, USB3 card, TPM, serial
Slot4	PCIe 3.0 x16	75 W slot	Third GPU or high-bandwidth card
Slot5	PCIe 2.0 x4	25 W	PERC / RAID (H710 in B1GMB42)
Slot6	PCI 2.3 (32-bit)	—	Legacy PCI via riser (Slot4 area)

Manual allows up to **two** full-height full-length PCIe x16 cards at 225 W each; B1GMB42 exceeds that count with three 75 W FirePro cards — confirm PSU label (**425 W** or **685 W**).

Serial PCIe add-in card (if installed): valid slots **1, 3, 4, 5** per Dell Serial PCIe Setup Guide.

Every external cable / port (outside the chassis)

Front panel (left to right, Fig 1)

#	Connector	Type	Connect
1	Microphone	3.5 mm TRS	Headset mic
2	Power button	—	— (chassis switch)
3	HDD activity LED	—	— (status only)
4	USB	USB 3.0	Keyboard, stick, hub
5–6	USB	USB 2.0 ×2	Mouse, audio, etc.
6–9	Optical drive	Slim SATA ODD ×2 (optional)	Install media
10	Headphone	3.5 mm TRS	Headphones
11	Line-in / mic	3.5 mm TRRS	Analog audio in

Rear panel (left to right, Fig 1)

#	Connector	Type	Connect
12	Serial	9-pin DB-9	Serial device (COM1 default)
13	USB	USB 2.0 ×2	Legacy peripherals
14	PS/2	Keyboard	PS/2 keyboard
15	USB	USB 3.0 ×2	Fast peripherals
16	Line-out	3.5 mm TRS	Powered speakers / amp
17	Security cable	Kensington slot	Physical lock
18	Padlock ring	—	Padlock hasp
19	Network	RJ-45 (Intel I217-LM)	Ethernet cable

#	Connector	Type	Connect
20	PS/2	Mouse	PS/2 mouse
21	USB	USB 3.0	Fast peripheral
22	USB	USB 2.0	Peripheral
23	Expansion slots	PCIe/PCI bracket	GPU DP/HDMI , PERC SAS, serial
24	Mechanical slot	Vent / blank	No active ports
25	AC power	IEC C14 inlet	Detachable AC cord to wall
26	PSU latch	—	Release PSU module

Rear GPU display ports (B1GMB42 as-built)

Each FirePro exposes **4× mini-DisplayPort** on the bracket. Plug the monitor into **whichever GPU** should drive the desktop (unpinned policy). Unused GPUs may stay headless.

Card	Bracket location	Video out
W5000 Slot1	Upper slot region	4× mini-DP
W5100 Slot2	Mid region	4× mini-DP
W5000 Slot4	Lower region	4× mini-DP

Every internal cable / header / slot (inside the chassis)

Chassis regions (Fig 2-3)

Region	What connects here
CPU heatsink + fan	5-pin CPU_FAN header on motherboard
Memory shrouds	DIMM1-8 under shrouds (B1GMB42: DIMM1/2/3 populated)
5.25" bay	Full-height ODD or 3.5" adapter / card reader
Slim ODD bays	Up to 2 slim SATA optical drives
Primary HDD cages	HDD0-HDD3 SATA + power (3.5" or 2.5" caddies)
PSU cable shroud	Routed SATA power, 24-pin, CPU 10-pin , drive harness
PCIe retention cage	Full-height cards (GPUs, PERC)
Internal speaker	4-pin speaker header
PSU module	Bottom/rear; slide-latch removal

Motherboard labeled connectors (Fig 4 — all 30)

Full numbered list is left of the Fig 4 photo in the PDF; summary sits under the photo. Expanded reference:

#	Dell label	Header / slot	Cable / device
1	PCI slot	Slot6 legacy PCI	PCI riser card
2	PCIe x16 (x4 wired)	Slot5	PERC H710 (B1GMB42)
3	PCIe 3.0 x16	Slot4	FirePro W5000 #2
4	PCIe 2.0 x1	Slot3	Empty — NIC, USB3, serial
5	PCIe 3.0 x16	Slot2	FirePro W5100
6	PCIe 3.0 x16 (x8 wired)	Slot1	FirePro W5000 #1
7	DIMM slots	Channel A	DDR4 RDIMM (DIMM1, DIMM5, ...)
8	Intrusion switch	2-pin	Chassis intrusion cable
9	CPU heatsink fan	5-pin	CPU fan tach/power
10	Processor socket	LGA2011-3	Xeon E5-1660 v3
11	DIMM slots	Channel B	DDR4 RDIMM (DIMM2, DIMM3, ...)
12	Front-panel audio	2×5 HD Audio	Front mic/headphone jack cable
13	Internal USB 2.0	2×5 header	Type-A internal port or cable

#	Dell label	Header / slot	Cable / device
14	Coin-cell battery	CR2032	BIOS clock / CMOS
15	Optional HDD fan	5-pin	Bay cooling fan
16	System fan	4-pin	Chassis fan #1
17	HDD temp sensor	2-pin	Thermal probe on drive cage
18	System fan	4-pin	Chassis fan #2
19	PWR_REMOTE	2-wire	Teradici / dual-host card (optional)
20	Thunderbolt sideband	5-pin	Dell Thunderbolt PCIe card (optional)
21	System fan	4-pin	Chassis fan #3
22	Password jumper	2-pin	BIOS setup password enable
23	Front panel + USB2	2×14	Power LED, HDD LED, power switch, USB2 front
24	Internal speaker	4-pin	Chassis beeper
25	USB 3.0 front panel	20-pin	Front USB3 port cable
26	Internal USB flexbay	2×5	Flex bay / media reader USB
27	SATA	7-pin ×6	HDD0-3 + ODD0-1 (see table below)
28	RTC_RST jumper	2-pin	Clear CMOS (with battery out)
29	24-pin ATX	Main power	From PSU — always required
30	CPU power	10-pin EPS	From PSU — always required

SATA port map (motherboard silkscreen □ device)

SATA label	Controller	Typical bay	B1GMB42 as-built
SATA3-HDD0	Intel AHCI	Primary internal bay 0	Likely Hitachi (sdb, ZFS)
SATA3-HDD1	Intel AHCI	Primary internal bay 1	Likely Seagate (sdc, Windows)
SATA2-HDD2	Intel AHCI	Primary internal bay 2	Empty or DVD
SATA2-HDD3	Intel AHCI	Primary internal bay 3	Empty
SATA2-ODD0	Intel AHCI	Slim ODD bay 0	PLDS DVD if installed (sr0)
SATA2-ODD1	Intel AHCI	Slim ODD bay 1	Empty

TEAM SSD (sda) is on a separate port (often flex bay or HDD1) and serves as **ZFS special vdev** — confirm with `lsblk` / cable trace after opening cover.

Internal drive power (from PSU)

Each 3.5"/2.5" device needs:

- **SATA data** □ motherboard port 27 (one of HDD0-3 / ODD0-1)
- **SATA power** □ PSU flat connector (15-pin SATA power)

Optical drives use the same SATA data + power pair. Do not mix data cables across ODD vs HDD ports if BIOS boot order matters.

Fan headers summary

Header	Purpose
CPU_FAN (9)	Stock CPU heatsink fan — required
SYS_FAN (16, 18, 21)	Chassis exhaust/intake fans
HDD_FAN (15)	Optional drive-bay fan
Memory shroud fans	Some configs — BIOS alerts if missing

Front-panel signal cable (item 23)

Single harness carries: power switch, power LED, HDD activity LED, and front USB2 ports. A disconnected harness causes "**Front I/O Cable failure**" at boot.

Optional / factory expansion wiring

Option	Internal connection
Dell Thunderbolt PCIe card	Slot3 or Slot5 + TB sideband (20)
Dual/Quad host PCIe card	Any x16/x8 slot + PWR_REMOTE (19) 2-wire to system board
Serial PCIe card	Slots 1, 3, 4, or 5 + rear DB-9 bracket
Media card reader	5.25" bay + flexbay USB (26) + SATA power
PERC H710	Slot5 + SAS cables to backplane (if cabled)

Storage — SATA / SAS / USB

Internal SATA disks

OS dev	Size	Model	Role
sda	238 GB	TEAM T253X6256G	rpool special vdev (~509 MB/s seq read)
sdb	1.4 TB	Hitachi HDS723015BLA642	rpool + bpool, EFI, plain swap (~145 MB/s)
sd c	466 GB	Seagate ST500DM002	exFAT / NTFS Windows + Ventoy partitions (dual-boot data)

ZFS pools (current)

Pool	Device	Health	Notes
rpool	Hitachi sdb4	ONLINE	Root + /home, encryption off ,
rpool special	TEAM sda	ONLINE	compression=lz4 Metadata + blocks <=32K;
bpool	Hitachi sdb2	ONLINE	spe- cial_small_blocks=32K /boot

SAS / RAID (Slot5)

Item	Detail
Card	PERC H710 @ 07:00.0
Driver	megaraid_sas
Status	FAULT — FW in FAULT state, Fault code:0x40000
Disks exposed	None

USB-attached storage

OS dev	Size	Model	Notes
sdd	30 GB	PNY USB (Ventoy)	Install/recovery stick — not production storage

USB (motherboard)

Controllers

Controller	Type	Typical ports
00:14.0 xHCI	USB 3.0	Rear USB3
00:1d.0 EHCI	USB 2.0	Rear/front USB2
00:1a.0 EHCI	USB 2.0	Internal headers / front

Attached devices (current)

Device	Notes
Realtek RTS5182	Front-panel card reader (empty slots)
CSR Bluetooth	Onboard BT dongle
C-Media USB audio	USB sound
Dell keyboard / wireless mouse	Input

Removed since pre-crash inventory: JMicron USB 2×2TB WD enclosures (were sdg/sdh, caused I/O errors — powered off).

Network

Port	Chip	Bus	Status
Onboard RJ-45	Intel I217-LM	00:19.0	Present
PCIe NIC	—	—	None installed
USB Wi-Fi	—	—	Not attached now (D-Link DWA-121 was on old scan)

Audio

Source	Device	Status
Motherboard	Intel C610 HD Audio 00:1b.0	Onboard analog jacks
GPU ×3	AMD HDMI audio on each FirePro	On display outputs
USB	C-Media USB audio	Attached

Memory (DIMM slots)

Slot	Status	Module
DIMM1	Occupied	4 GB Hynix DDR4-2133
DIMM2	Occupied	4 GB Hynix DDR4-2133
DIMM3	Occupied	4 GB Hynix DDR4-2133
DIMM5	Empty	—
DIMM6	Empty	—
DIMM7	Empty	—

Total: ~15 GiB usable. Board has additional slots — target 32 GB+.

SATA controllers

Controller	Ports
00:1f.2 6-port SATA (AHCI)	SATA 0-5
00:11.4 sSATA (AHCI)	sSATA 0-3

DVD/ODD and exact bay silkscreen mapping not captured in software inventory.

Headless GPU use

GPUs do **not** need a monitor to run Vulkan/OpenGL compute or offscreen render. Only the **desktop session** GPU needs a cable (or dummy plug).

Use case	Headless OK?
vkcube / Vulkan stress (./gpu-stress.sh)	Yes
OpenCL / offscreen GL on W5000/W5100	Usually yes
ROCm / HIP ML	No — W5000/W5100 not in ROCm matrix
GNOME Wayland desktop	One GPU needs display or dummy plug

```
lspci | grep -i vga
ls -l /dev/dri/
DRI_PRIME=1 glxinfo | grep 'OpenGL renderer' # pick non-primary GPU
```

Video Manual (unpinned multi-GPU)

Policy: No forced primary GPU. Plug the monitor into whichever card should drive the desktop.

Shared limits

Limit	Value
AMD rated ports per GPU	4 mini-DP
Physical ports installed	12 (4 per card)
Wayland desktop GPUs	1 (session binds to one GPU)
X11 multi-GPU displays	~12 with xrandr providers

Config files (etc/)

File	Purpose
etc/modprobe.d/amdgpu-multigpu.conf	runpm=0 — stable multi-GPU
etc/udev/rules.d/99-amdgpu-multigpu.rules	GPU tags (no setpci pin)
etc/X11/xorg.conf.d/20-amdgpu-multi-gpu.conf	X11 multi-GPU, no BusID
etc/gdm3/custom.conf	WaylandEnable=true
etc/environment.d/99-amdgpu-wayland.conf	Mesa radeonsi
etc/apply.sh	Installer — run sudo ./etc/apply.sh

Wayland session

- GDM gear menu: **Ubuntu** (Wayland) or **Ubuntu on Xorg**
- Verify: `echo $XDG_SESSION_TYPE` should print wayland
- Headless GPUs: use **HDMI/DP dummy plugs** if an app needs a CRTC

X11 session

- Multi-GPU: `xrandr --listproviders` then per-provider `xrandr`
- Dual-boot Windows: set primary display in Windows separately

W5000 vs W5100 as display GPU

No meaningful desktop difference. W5000 slightly stronger for 3D on the display GPU.

Troubleshooting

Symptom	Likely cause	Fix
Boot failure after disk swap	TPM + ZFS encryption on old install	This reinstall avoids that — keep encryption off unless TPM is configured first
UI freezes, high load	ZFS on HDD, low RAM	Add TEAM special vdev; add RAM; avoid swap thrash
Cannot find any crtc	Headless GPU	Expected; dummy plug if needed
PERC no disks	FW fault 0x40000	PERC BIOS / battery / replace or IT-mode flash
Wrong display GPU	Cable on different card	Move cable (unpinned policy)
<code>show_signal_msg: N</code> <code>callbacks suppressed</code>	Kernel rate-limiting duplicate logs	Not N separate errors — read the one line above

Triple-GPU / PSU

- **Installed:** 2× W5000 + 1× W5100 (75 W each, slot-powered)
 - **Estimated peak:** ~440 W — confirm PSU label (**425 W** tight, **685 W** comfortable)
 - Stress test: `./gpu-stress.sh 60 vkcube`
-

Summary — Occupied vs Free

Resource	Used	Free
PCIe Slot1 (x8)	FirePro W5000	—
PCIe Slot2 (x16)	FirePro W5100	—
PCIe Slot3 (x1)	—	Open (NIC, USB3, serial)
PCIe Slot4 (x16)	FirePro W5000	—
PCIe Slot5 (x4)	PERC H710 (faulted)	—
PCIe Slot6 (PCI)	Riser only	Legacy PCI card
SATA HDD0–3	Hitachi + Seagate (+ TEAM)	1–2 bays

Resource	Used	Free
SATA ODD0-1	DVD (if installed)	1 slim bay
DIMM (8 slots)	3 × 4 GB	5 slots
Front USB	All active	—
Rear USB	Most active	Some ports
Internal USB headers	Card reader, BT	1-2 headers
Fan headers	CPU + system	1-2 optional
ZFS special vdev	TEAM SSD	—
Encryption	off	—

IndianaDell workspace

```
IndianaDell/
├── B1GMB42-slot-port-inventory.md    # this file
├── blgmb42.report / B1GMB42.ioperf  # live inventory + disk benchmark
├── 8SNZK02.report / 8SNZK02.ioperf  # second machine (thumper / Titan Xp)
├── dellmerge.sh, iotest.sh, gpu-stress.sh
├── etc/                             # multi-GPU Wayland/X11 config
├── amd-radeon/                     # manual ROCm install toolkit
├── FactoryDocs/                   # Dell support packages (sorted)
│   ├── _incoming/                 # drop new downloads here
│   ├── _sort_factory_docs.py      # dedupe + sort script
│   └── README.md                  # folder index
```

FactoryDocs ingest

```
cp ~/Downloads/* ~/Documents/IndianaDell/FactoryDocs/_incoming/
python3 ~/Documents/IndianaDell/FactoryDocs/_sort_factory_docs.py
```

Pre-crash: 101 sorted Dell packages (~500 MB). **Now:** 19 on disk. See FactoryDocs/MANIFEST-pre-crash.txt for the full list of what to re-download (GPU, PERC, audio, TPM, Win7/10 CABs, SSD firmware, etc.).

Action Items

1. **Apply video config** (if not done): `cd ~/Documents/IndianaDell && sudo ./etc/apply.sh` then reboot.
2. **RAM:** Upgrade to 32 GB+ when possible (currently 3×4 GB).
3. **PERC H710:** Fix firmware fault or replace before using SAS bays.
4. **PSU:** Confirm wattage on PSU label.
5. **Do not re-enable ZFS encryption** until TPM + recovery strategy is documented.
6. **FactoryDocs:** Re-grab any drivers that failed mid-download (GPU, audio, encryption zips).

Refresh commands

```
cd ~/Documents/IndianaDell
chmod +x dellmerge.sh gpu-stress.sh etc/apply.sh
sudo bash ./dellmerge.sh > blgmb42.report
pandoc B1GMB42-slot-port-inventory.md -o B1GMB42-slot-port-inventory.pdf \
  --pdf-engine=xelatex -V mainfont="Noto Sans" -V monofont="DejaVu Sans Mono"
zpool status && zpool list -v rpool
lsblk -o NAME,SIZE,MODEL,TRAN,FSTYPE,MOUNTPOINT
lspci -nn | grep -iE 'vga|raid'
sudo dmesg | grep -iE 'megaraid|amdgpu|fault'
```

